

ReadMe Document

This ReadMe document will help you with a quick navigation through the contents of the Pre-Reading material, Preview videos, and Individual Labs for the upcoming session.

Outcomes

After completion of this week's sessions, & activity participants will be able to:

1. get the higher-level concept of Backpropagation & its applications
2. get the higher-level concept of the Convolution layer & its applications
3. understand and use the PyTorch for building ML/ DL model
4. understand & implement the details of the CNN
5. understand the details of back propagation
6. understand & implement the concept of encoder- decoder and auto-encoder architecture

Lecture Topics

1. Convolutional Layer, Backpropagation
2. Pytorch
3. CNN, Back Propagation and Auto-encoder

Preview Videos

The Preview Videos Link for the respective sessions are available a week in advance (i.e. every Monday), to help you understand the concepts that will be covered in the upcoming lectures. There could be one or more videos depending on the lecture details. The video/videos should be viewed before attending the lecture for better understanding.

There are 5 preview videos for this session and they should be viewed as given below:

1. Convolutional Layer : This is a 30-minute video. From this video you will learn about Convolutional layer, Zero padding, Stride and Fully Connected layer.
2. Backpropagation : This is a 35-minute video. This video will help you to understand how backpropagation adjusts the weights in a neural network, to achieve minimization in loss. It will help in understanding how the chain rule method is used in backpropagation.
3. Pytorch: This is a 8-minute 21 sec video. From this video, you will understand the basics of Pytorch
4. Neural Networks For Dimensionality Reduction: This is a 3-minute video. This video helps you to understand how a neural network helps in Dimensionality reduction
5. Convolutional Neural Network: This is a 2-minute video. From this video, you will understand how to classify the images using CNN. Imagenet, Traditional Machine Learning Approach and Deep Learning Approach.

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Pre Reading Material

The pre-reading material is given on (1) Convolutional Layer (2) Backpropagation. You should read the material before you attend the lecture. This pre-reading will provide the gist of the lecture and facilitate your understanding during the lecture.

Reference Material

The Reference Material is given on (1) Convolutional Layer (2) Backpropagation. You should read the material after attending the lecture. This reference material will provide further depth into the topic.

Individual Labs

1. Convolutional Layer
2. Backpropagation
3. Pytorch